Jose Alberto Tupayachi Silva

AI & Process Automation Engineer · **Data, ML & Agentic Workflows**

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SUMMARY

Industrial & Systems Engineer and AI/data engineer with 6+ years automating business and engineering processes end to end, from manual and expert-driven work to production pipelines, agentic LLM workflows and self-service dashboards. Has cut expert review cycles from **days to under 1 hour**, moved an **18M-customer** Oracle warehouse to an AWS lakehouse with **zero data loss**, and automated CI/CD, data-quality checks, scoring and invoice verification across energy, finance, consulting and national-lab teams. Combines operations research, simulation and reliability engineering with ML, applied to freight logistics, predictive maintenance and utility operations. Led the AI/data workstream of a **$3M** DOE ARPA-E project, mentors junior students, and works in Agile sprints with business and technical stakeholders.

TECHNICAL SKILLS

Process Automation & Orchestration: End-to-end workflow automation, agentic LLM workflows (LangGraph, tool calling), Airflow, Prefect, Databricks Workflows, Jenkins CI/CD, Python and Bash/Shell scripting, REST/GraphQL and Salesforce API integrations, automated data-quality checks, monitoring and alerting, document and report generation, desktop automation (PyQt), batch parallelization (Slurm, GNU Parallel)

Industrial Engineering & Operations Research: Linear, integer, network-flow and stochastic programming (Gurobi, CPLEX), discrete-event simulation (AnyLogic), reliability engineering (Weibull, FMEA, remaining-useful-life prognostics), condition monitoring and fault detection, supply chain and intermodal logistics, engineering economic analysis, KPI design, usability testing, Agile/Scrum, requirements gathering with business stakeholders

AI & Machine Learning: Machine learning engineering, deep learning, LLMs, RAG, agentic AI workflows, CNNs, 1D-CNN, Vision Transformers (ViT, Swin), Graph Neural Networks, TensorFlow, PyTorch, statistical and predictive modeling, clustering, regression, tree-based and boosting models (Random Forest, LightGBM, XGBoost), model tuning and calibration, explainable AI

LLM & Agentic Frameworks: LangChain, LangGraph, LiteLLM, vLLM, Ollama, llama.cpp, QLoRA fine-tuning, BERT and sentence transformers, vector databases, knowledge graphs (Neo4j), ontology-guided reasoning

Data Engineering: Apache Spark (Scala/PySpark), Hadoop/MapReduce, Hive, Kafka, ETL/ELT, batch and real-time streaming, Delta Lake, Databricks Workflows, Unity Catalog, dbt, Airflow, Prefect, data modeling, entity resolution, data quality, anomaly detection, Snowflake, Amazon Redshift, dimensional modeling, performance optimization

Cloud & Infrastructure: AWS (S3, EMR, Redshift, SageMaker, ECS), Microsoft Azure, Google Cloud (BigQuery), Docker, Kubernetes, UNIX/Linux, KVM, Slurm and GNU Parallel (HPC), CI/CD, Jenkins, Git, GitLab

Databases: PostgreSQL, MySQL, SQL Server, MongoDB, Cassandra, Neo4j

Languages: Python, SQL, Scala, Java, R, SAS, STATA, SPSS, Bash, Dart, PHP, C++, CUDA

Web, Mobile & APIs: Django, Flask, Streamlit, Dash, React, Flutter, Laravel, GetX, REST and GraphQL APIs, Power BI, Tableau, Looker Studio

Geospatial & Networks: NetworkX, WMS, GeoServer, OverPass API, Turf.js

Certifications: Microsoft Fabric DP-600 and DP-700, MS Transact-SQL (T-SQL), AWS, Google Cloud

EXPERIENCE

Developer · Oak Ridge National Laboratory — Mar 2025 – Present
Oak Ridge, TN

- **Automated manual image inspection:** built a multi-pathway **Vision Transformer** agentic pipeline that classifies microstructure, defects and phases across **1,000+ SEM images**, replacing days-per-batch expert annotation with reproducible classification and auto-generated structured reports.
- **Automated document review:** deployed a **LangChain/LangGraph RAG** workflow (Qwen and Llama on Ollama, vector DB) that ingests and cross-references **200+ technical papers**, cutting expert review from **2–5 days to under 1 hour** per batch with full source traceability.
- **Automated data QA:** fine-tuned a **Swin Transformer** that quality-screens wind speed and direction from distributed weather stations before they feed atmospheric plume models, and built explainable ML (DL, 1D-CNN) for extreme-weather prediction.
- **Automated dashboard creation:** built a natural-language geospatial dashboard generator with knowledge-graph reasoning and agent self-validation (llama.cpp, LangChain, vLLM), so non-experts build analyses without engineering help; **filed as an ORNL IP disclosure (2026)** and shipped as a public Docker image.
- Deployed a GPU-accelerated (cuML) spectral and spatial analysis pipeline that processes raw IONTOF ToF-SIMS instrument files (.itax/.itmx) end to end.

Graduate Research Assistant · University of Tennessee, Knoxville — Aug 2022 – Present
Knoxville, TN

- Served as **Lead AI/Data Engineer** on the **$3M DOE ARPA-E RECOIL** freight digital twin: owned the AI/data workstream across **100+ intermodal nodes**, automating route optimization, demand forecasting and what-if analysis behind a production LLM chatbot that cut stakeholder decision turnaround from **hours to minutes**.

- Parallelized and scheduled experiment runs on HPC (**Slurm, GNU Parallel**) and **mentored junior students** in model development and Git workflows to cut run times; led a Git workshop for a research-methods class.
- Shipped Flutter mobile health apps for **Android, iOS, watchOS and Wear OS** (SmartShots, Active Caregiver's Toolkit) published on the App Store and Google Play; SmartShots won the IISE DAIS Student Mobile App Competition.
- Developed **decentralized (federated) prognostics** for industrial equipment, detecting faults and estimating remaining useful life across distributed assets with heterogeneous sensor data, without centralizing raw data (*Computers & Electrical Engineering*, 2026).
- Built an ontology-guided intermodal routing optimizer (linear optimization) and a Traffic-Weather GCN for EV charging-demand forecasting (**MAE 0.064**, more than **10x** lower than classical baselines).

Data Engineer · Indra Mar 2022 – Aug 2022

- Led the migration of **3 core tables of about 5,000 variables each (about 18M customers)** from a legacy Oracle warehouse to a partitioned AWS lakehouse (S3, EMR, Hive) with **zero data loss**, redesigning schemas into dimensional star models and re-implementing Oracle stored-procedure logic in PySpark/Scala.
- Tuned Spark executor memory and cores per job, broadcast-join thresholds, partition pruning and file coalescing, removing OOM shuffle spills and small-file problems that slowed downstream queries.
- Automated entity resolution and deduplication across the 18M-record client base, and built scheduled ETL/ELT pipelines in Python, Scala and Shell on UNIX/Linux in Agile cycles.
- Automated CI/CD with Jenkins (builds, unit tests, live deployment, rollback, backups) to cut manual effort and deployment risk.
- Built an automated data-quality engine (profiling, validation rules, drift detection, monitoring and alerting) to catch data issues before downstream reporting.

Business Intelligence Analyst · Globokas Jan 2022 – Mar 2022

- Automated KPI reporting by integrating **10+ heterogeneous sources** (transactional databases, collections, billing, payment-center feeds) into unified BigQuery, Looker Studio and Power BI pipelines that feed self-refreshing executive KPI dashboards for multiple business clients.

Data Analyst · Enel Distribución Nov 2020 – Dec 2021

- Segmented **1.6M utility customers** with K-Means and hierarchical clustering on payment history and demographics, lifting collection-campaign hit rate by **about 15%** and lowering cost per recovered account.
- **Automated invoice verification** and special-case processing with a PyQt desktop app, removing manual verification steps from the billing workflow.
- Built Power BI and Tableau operational dashboards and managed SQL/T-SQL databases and API-based Salesforce integrations.

Developer, Portfolio Valuation · GPS Management Aug 2020 – Nov 2020

- **Automated the loan-scoring workflow** with machine learning models for non-performing loan (NPL) recovery and portfolio valuation, reducing manual processing time (presented at POMS Lima, 2022).

IT Intern · Quiron Salud Jan 2019 – Aug 2019

- Maintained healthcare plan information systems and supported electronic health record integration and database administration for clinical operations.

PROJECTS

RECOIL · Cognitive Freight Transportation Digital Twin Jul 2024 – Mar 2026
Lead AI/Data Engineer · DOE ARPA-E, $3M (DE-AR0001780) · m1.recoil.ise.utk.edu

- Modeled a national U.S. freight network of **100+ intermodal nodes** (road, rail, waterway, EV charging); the ontology-guided GIS optimizer projects a **15–20% cut in per-route CO_2** versus truck-only baselines (arXiv:2410.18089).
- Built and published the Traffic-Weather GCN on proprietary ChargePoint data (**86 stations, 13 months**): **MAE 0.064, RMSE 0.573**, more than **10x** lower error than XGBoost, Random Forest, LightGBM and SVM baselines (arXiv:2510.09048).
- Deployed a trade-off analysis chatbot (Neo4j knowledge graph, vector embeddings, BERT/sentence-transformer literature index) used by **20+ domain experts and DOE stakeholders**, shortening decision turnaround from **hours to minutes** for non-technical users.
- Automated a real-time streaming feedback loop that continuously re-optimizes routing under weather, traffic and demand changes, reducing unnecessary truck miles versus static batch baselines.

Automated Geospatial Analysis Interface · ORNL, NSRD Mar 2025 – Present
github.com/jtupayachi/smartgiscavp · ORNL IP disclosure (2026)

- End-to-end LLM interface that lets non-expert users query, process and visualize spatial datasets in natural language, with agent self-validation; a **dual-LLM pipeline** (a planner model hands off to a coder model) plans, generates, builds and serves multi-page React apps live from a plain-language request.
- Published **3 Docker Hub images with 1,590+ total pulls** (LLM OCR pipeline and UI: **1,550+ pulls**), **7 fine-tuned models, 5 datasets and 1 Space** on Hugging Face, and **41 public GitHub repositories**.

SmartShots · Tennessee Dept. of Health Dec 2023 – Dec 2024
Full-Stack Developer · Winner, IISE DAIS Student Mobile App Competition 2024

- Built the Laravel backend and Flutter app (real-time notifications, nearby vaccination-provider lookup) to improve childhood vaccination access across Tennessee; published on the App Store and Google Play and documented in *Health Informatics Journal* (2026).

Active Caregiver's Toolkit (ACTAPP) · Hillman Emergent Innovation (HEI) Jul 2024 – Present
Mobile Developer

- Built a Flutter (Dart 3, GetX) physical-activity app for rural Appalachian caregivers with an integrated FAQ chatbot that logs interaction, navigation and engagement metrics, feeding the study's behavioral-outcome analysis under IRB protocols with no PII retained.

LEADERSHIP & MENTORSHIP

- **Lead AI/Data Engineer** of the $3M DOE ARPA-E RECOIL project (Jul 2024 – Mar 2026): owned the AI and data workstream (optimization, forecasting, LLM chatbot, streaming) and presented the production platform to DOE program sponsors.
- **Mentored junior students** in model development, GitHub and GNU Parallel for faster execution.
- **Teaching assistant** for 3 UTK Industrial Engineering courses (simulation, reliability, engineering management), **guest lecturer** on digital twins at the University of Central Florida (with Prof. Soheil Sabri) and **workshop leader** on Git for a UTK research-methods class.
- **Senator**, UTK Graduate Student Senate, Industrial and Systems Engineering (2025): represented students and advocated for academic and professional development opportunities.
- **Peer reviewer** for 13 journals and conferences, including IISE Transactions, PLOS ONE, Software: Practice and Experience and the Winter Simulation Conference.

EDUCATION

M.S., Industrial Engineering · University of Tennessee, Knoxville Aug 2022 – May 2024
GPA 3.93 · Coursework: optimization methods, integer programming, network flow optimization, stochastic programming, advanced systems simulation, applied data science, usability testing, strategic management of technology

B.S., Industrial Engineering · Pontifical Catholic University of Peru Mar 2014 – May 2019

AWARDS & HONORS

- **IISE Future Faculty Fellow** (2025–2026), USA
- **Best Paper Award**, CIE51, 51st Conference on Computers and Industrial Engineering, Sydney, Australia (2024)
- **Winner, IISE DAIS Student Mobile App Competition**, SmartShots project, Montreal (2024)
- **HIDA Helmholtz Visiting Researcher** award, Karlsruhe Institute of Technology, Germany (2024)
- **Holiday Fellowship**, University of Tennessee, Knoxville (2022, 2023, 2024)

SELECTED PUBLICATIONS 373 citations · h-index 7 · i10-index 5 (Google Scholar)

- **Tupayachi, J.**, Khan, A. N., & Li, X. (2026). Scalable decentralized prognostics for industrial systems under data heterogeneity. *Computers & Electrical Engineering, 133*, 111023.
- Khan, A. N., **Tupayachi, J.**, Dabbs, W., Chamberlin, S., Yao, B., & Li, X. (2026). CliniSense AI for automated clinical skills assessment with real-time feedback. *npj Digital Medicine*.
- Xu, H., Sun, Y., **Tupayachi, J.**, Omitaomu, O., Zlatanova, S., & Li, X. (2026). Towards autonomous freight intermodal optimization via generative AI and agentic digital twins. *Computers & Industrial Engineering*, 112464.
- Xu, H., Li, X., **Tupayachi, J.**, Lian, J. J., & Omitaomu, O. A. (2024). Automating bibliometric analysis with sentence transformers and RAG. *ACM SIGSPATIAL Urban-AI Workshop*, 1–10.
- *Also:* 3 more journal papers (Smart Cities, Health Informatics Journal, Drones), 2 conference papers, 4 pre-prints, and talks at IISE, INFORMS, ACM SIGSPATIAL and the ANS Annual Meeting. More: Google Scholar.

REFERENCES

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